EXACT 2026

Kick-off Workshop

The 2nd International XAI Challenge for Transparent Educational Question-Answering

IEEE IJCNN 2026 Competition

EXACT 2026 Organizing Committee

Ho Chi Minh City University of Technology (HCMUT), VNU-HCM, Vietnam

May 9, 2026

Table of Contents

1. Program Introduction
2. Opening Remarks
3. Competition Overview & Guidelines
4. Dataset Type 1: Logic-Based Educational Queries
5. Dataset Type 2: Physics Problems
6. Suggested Approaches & Strategies
7. Technical Demo

Table of Contents

1. Program Introduction
2. Opening Remarks
3. Competition Overview & Guidelines
4. Dataset Type 1: Logic-Based Educational Queries
5. Dataset Type 2: Physics Problems
6. Suggested Approaches & Strategies
7. Technical Demo

Workshop Agenda

Today's workshop is organized into the following sessions:

1. **Opening Remarks** — *Welcome & introduction*
2. **Competition Overview & Guidelines** — *Timeline, rules, evaluation*
3. **Dataset Type 1: Logic-Based Educational Queries** — *Format & examples*
4. **Dataset Type 2: Physics Problems** — *Format & examples*
5. **Suggested Approaches & Strategies** — *Practical guidance*
6. **Technical Demo** — *API, pipeline, submission*
7. **Q&A Session** — *Open discussion*

Table of Contents

1. Program Introduction
2. Opening Remarks
3. Competition Overview & Guidelines
4. Dataset Type 1: Logic-Based Educational Queries
5. Dataset Type 2: Physics Problems
6. Suggested Approaches & Strategies
7. Technical Demo

Welcome to EXACT 2026

EXACT 2026 is the 2nd International XAI Challenge for Transparent Educational Question-Answering.

Co-organized by:

- Ho Chi Minh City University of Technology (HCMUT), VNU-HCM, Vietnam
- University of Naples Parthenope, Italy

Held under:

- IEEE International Joint Conference on Neural Networks (IJCNN) 2026
- IEEE World Congress on Computational Intelligence (WCCI 2026), Maastricht, The Netherlands

Previous Edition

EXACT 2025 at IEEE IJCNN 2025 attracted 107 participants across 30 teams from multiple countries, producing the first public benchmark for explainable academic-regulation QA.

What's New in 2026

- Expanded into STEM reasoning (Physics)
- Upgraded evaluation criteria
- Structured explanation assessment

International Challenge Chairs

The quality and fairness of EXACT 2026 are overseen by a distinguished international committee:

Name	Affiliation	Country
Prof. Quan Thanh Tho	Ho Chi Minh City University of Technology (HCMUT)	Vietnam
Prof. Emanuel Di Nardo	University of Naples Parthenope	Italy
Prof. Nguyen Duc Anh	University of South-Eastern Norway	Norway
Prof. Fabien Baldacci	University of Bordeaux	France
Prof. Nguyen Le Minh	Japan Advanced Institute of Science and Technology	Japan

Table of Contents

1. Program Introduction
2. Opening Remarks
3. Competition Overview & Guidelines
4. Dataset Type 1: Logic-Based Educational Queries
5. Dataset Type 2: Physics Problems
6. Suggested Approaches & Strategies
7. Technical Demo

Competition Timeline

#	Date	Milestone
1	Apr 10 to May 10, 2026	Team registration
2	May 9, 2026	**Kick-off Workshop & dataset release** *(today)*
3	May 10 to May 30, 2026	Main competition phase
4	Jun 1 to Jun 2, 2026	Phase 1 evaluation & feedback
5	Jun 3 to Jun 4, 2026	Model refinement period
6	Jun 5 to Jun 7, 2026	Phase 2 evaluation
7	Jun 10, 2026	Top 10 announcement
8	Jun 15, 2026	Public Test Day (live demo before jury)
9	Jun 30 to Jul 15, 2026	Paper submission (Top 10 teams)
10	Nov 16 to 18, 2026	Presentation at CSoNet 2026

Eligibility & Team Formation

Who Can Participate?

- High school students
- Undergraduates and graduate students
- Engineers and researchers
- **No restriction** on age, nationality, or affiliation

Team Rules

- Up to **5 members** per team
- Interdisciplinary and cross-institutional teams are encouraged
- Individuals who registered alone will be matched into teams by the organizers

Technical Requirements

To ensure fairness and reproducibility, all submissions must follow these rules:

✔ Allowed

- Open-source LLMs only
- Maximum **8 billion parameters** (applies to every LLM component)
- Symbolic reasoning tools (Z3 Solver, custom engines, etc.) are encouraged
- Any approach producing explainable results

✘ Prohibited

- Closed-source models (GPT, Claude, Gemini, etc.)
- Undisclosed external datasets for fine-tuning
- Submissions relying on proprietary APIs

⚠ Disclosure

All external datasets used for fine-tuning must be **fully disclosed** in the submission.

Evaluation Criteria

Submissions are assessed across three dimensions:

ID	Criterion	Description
P1	**Correctness**	Accurate and precise answers for the given queries
P2	**Explanation Quality**	Clear, coherent natural-language explanations that justify each answer
P3	**Reasoning Depth**	Supporting evidence such as FOL derivations, Chain-of-Thought steps, or structured proofs

Evaluation process:

1. **Phase 1 (Selection):** Submissions are scored automatically against ground-truth answers, then reviewed by the organizing committee for explanation quality. Detailed feedback is sent back to teams.
2. **Phase 2 (Selection):** Same evaluation pipeline as Phase 1 (auto correctness + committee review of explanations) after teams refine their models.

Submission Format

What to Submit

- A working **API endpoint** that accepts queries and returns answers with explanations
- A **one-page solution description** outlining your approach, models used, and any external data

API Requirements

- Must accept input queries in the specified JSON format
- Must return structured output (answer + explanation)
- Maximum **60 seconds** per request
- Detailed format will be provided with the dataset release

Awards & Publication Opportunities

Rank	Benefits
Top 5	Cash prizes + invited presentation at the 15th International Conference on Computational Data and Social Networks (CSoNet 2026), Nov 2026 at HCMUT. Papers published by Springer in LNCS/LNAI, indexed by Scopus, DBLP, Web of Science, ACM Digital Library.
Top 10	Invited to submit a paper to the Special Session "Explainable AI for Educational QA" at CSoNet 2026.
All valid	Official certificate from the Challenge Chairs and conference organizers.

Table of Contents

1. Program Introduction
2. Opening Remarks
3. Competition Overview & Guidelines
4. Dataset Type 1: Logic-Based Educational Queries
5. Dataset Type 2: Physics Problems
6. Suggested Approaches & Strategies
7. Technical Demo

Type 1: Problem Definition

Objective

Build a system that answers questions about **university academic regulations** (grading policies, enrollment rules, scholarship criteria) and provides **logical explanations** grounded in First-Order Logic (FOL).

Why is this challenging?

- Regulations contain complex conditional logic (if-then rules, exceptions, nested conditions)
- Answers must be provably correct, not just plausible
- Explanations must trace back to specific regulation clauses
- Requires bridging natural language understanding with formal logic

Type 1: Dataset Format

Statistics

411 records containing **808 questions** in total.
Question types: 358 Multiple Choice, 216 Yes/No/Uncertain, 234 Open-ended.
Coverage: 99% of records include FOL premises; 100% include human-written explanations.
Average per record: ∼10.8 NL premises, ∼1.97 questions.

Each record includes:

Field	Description
Natural-language premises	The regulation text describing rules and conditions
FOL annotations	First-Order Logic representation of the premises
Questions	One or more questions about the regulation
Ground-truth answers	Correct answers (Yes/No/Uncertain or specific values)
Human-written explanations	Step-by-step reasoning explaining why the answer is correct

Type 1: Example

Real sample from the dataset

Premises (NL):

1. Every student is understanding the material.
2. Every student is asking questions.
3. Every student is revising.

Premises (FOL): $\forall x\, \text{IsUnderstandingMaterial}(x)$; $\forall x\, \text{IsAskingQuestions}(x)$; $\forall x\, \text{IsRevising}(x)$

Question: "Do all students both understand the material and ask questions, according to the premises?"

Answer: Yes

Explanation: Premise 1 states all students understand the material, and premise 2 states all students ask questions, so all students both understand the material and ask questions.

Type 1: Key Challenges

Reasoning Challenges

- Multi-step logical inference across multiple regulation clauses
- Handling negation and exceptions
- Distinguishing between "No" and "Uncertain" when information is incomplete
- Numerical comparisons embedded in natural language

Explanation Challenges

- Generating FOL derivations that are correct and complete
- Producing human-readable explanations that match the formal proof
- Maintaining consistency between the answer and the explanation
- Citing specific premises used in reasoning

Table of Contents

1. Program Introduction
2. Opening Remarks
3. Competition Overview & Guidelines
4. Dataset Type 1: Logic-Based Educational Queries
5. Dataset Type 2: Physics Problems
6. Suggested Approaches & Strategies
7. Technical Demo

Type 2: Problem Definition

Objective

Build a system that solves **text-based physics problems** focusing on electric circuits and electrostatics, and provides **step-by-step Chain-of-Thought reasoning** for each solution.

Why is this challenging?

- Problems require multi-step numerical computation
- Must correctly identify which physics formulas to apply
- Intermediate steps must be shown and logically connected
- Final answers must include correct units
- Open-source LLMs (8B parameters or fewer) often struggle with precise arithmetic

Type 2: Dataset Format

Statistics

1,755 text-only physics problems covering electric circuits and electrostatics.
Average question length: 38.9 words (median 34).
Average CoT solution: 113 words across $\sim$4.7 reasoning steps.
Coverage: 77% include an explicit numerical final answer; 76% include the SI unit (55 distinct units overall, e.g. N, V/m, V, J, Ω, A, W, μF, nC).

Each problem includes:

Field	Description
Problem statement	A physics question described in natural language
Chain-of-Thought reasoning	Step-by-step solution showing formulas, substitutions, and intermediate results
Final answer	Numerical value
Unit	SI unit of the final answer (when applicable)

Type 2: Example

Real sample from the dataset (id: THCB078)

Problem: "A parallel circuit has $R_1 = 30\,\Omega$, $R_2 = 60\,\Omega$, $U = 12\,\text{V}$. Calculate the equivalent resistance."

Chain-of-Thought:

1. Apply the parallel formula: $\dfrac{R_1 \cdot R_2}{R_1 + R_2} = \dfrac{30 \cdot 60}{30 + 60} = \dfrac{1800}{90}$
2. Result: $20\,\Omega$

Answer: $R_{td} = 20.0$ **Unit:** Ω

Type 2: Key Challenges

Computation Challenges

- Multi-step arithmetic with fractions and decimals
- Correct formula selection from multiple candidates
- Unit conversions and consistency
- Handling edge cases (e.g., zero resistance, open circuits)

Explanation Challenges

- Generating coherent Chain-of-Thought that a human can verify
- Showing intermediate values, not just the final answer
- Matching the reasoning depth expected by the evaluation criteria (P3)
- Balancing detail with clarity

Table of Contents

1. Program Introduction
2. Opening Remarks
3. Competition Overview & Guidelines
4. Dataset Type 1: Logic-Based Educational Queries
5. Dataset Type 2: Physics Problems
6. Suggested Approaches & Strategies
7. Technical Demo

General Architecture

A typical system for EXACT 2026 may combine:

1. **Natural Language Understanding:** Parse the input query, identify question type (Type 1 or Type 2), extract key entities and constraints
2. **Reasoning Engine:** Apply logic-based or computation-based reasoning
   - Type 1: Symbolic reasoning (e.g., Z3 Solver), FOL theorem proving
   - Type 2: Equation solving, symbolic math, step-by-step computation
3. **Explanation Generation:** Produce natural-language explanations aligned with the formal reasoning steps
4. **Answer Formatting:** Structure the output in the required API format

Note: This is one possible architecture. Any approach that produces accurate, explainable results is welcome.

Approaches for Type 1 (Logic-Based)

Symbolic Reasoning

- Convert NL premises to FOL using an LLM
- Feed FOL into a solver (Z3, Prover9, custom engine)
- Use solver output to generate explanations
- **Pro:** Provably correct answers
- **Con:** NL-to-FOL conversion is error-prone

Neurosymbolic Hybrid

- Use LLM for understanding + symbolic engine for reasoning
- LLM generates candidate FOL, solver verifies
- Iterative refinement: if solver rejects, LLM re-generates
- **Pro:** Combines flexibility with rigor
- **Con:** More complex pipeline

Pure LLM with Prompting

Use Chain-of-Thought, few-shot examples, and self-consistency checks. Simpler to implement but may lack formal guarantees.

Approaches for Type 2 (Physics)

LLM + Code Execution

- LLM generates Python/SymPy code for computation
- Execute code to get precise numerical answers
- Use LLM to narrate the steps
- **Pro:** Precise arithmetic
- **Con:** Code generation errors

Retrieval-Augmented Generation

- Build a knowledge base of physics formulas and solved examples
- Retrieve similar problems as few-shot examples
- LLM solves with retrieved context
- **Pro:** Leverages existing solutions
- **Con:** Limited by retrieval quality

Practical Tips

- Fine-tune on the training data to adapt the LLM to the specific question format
- Use self-verification: have the model check its own answer by substituting back
- Ensemble multiple approaches and select the most consistent answer

Recommended Open-Source Models

Some open-source LLMs within the 8B parameter limit to consider:

Model	Parameters	Strengths
Llama 3.1 8B	8B	Strong general reasoning
Qwen 2.5 7B	7B	Multilingual, math capabilities
Phi-3 Mini	3.8B	Compact, strong reasoning for size
Mistral 7B	7B	Fast inference, good code generation
DeepSeek-R1-Distill	8B	Distilled reasoning capabilities

Note: Exact eligibility of borderline models (e.g., 8.19B) will be clarified during Q&A. The rule of thumb: nominal 8B-class models are accepted.

Table of Contents

1. Program Introduction
2. Opening Remarks
3. Competition Overview & Guidelines
4. Dataset Type 1: Logic-Based Educational Queries
5. Dataset Type 2: Physics Problems
6. Suggested Approaches & Strategies
7. Technical Demo

Technical Demo: Overview

In this demo, we will walk through:

1. **Dataset loading:** How to read and parse the provided dataset files
2. **Input/Output format:** The exact JSON schema for API requests and responses
3. **API usage:** How to set up and test your API endpoint
4. **End-to-end pipeline:** A simple baseline showing the full flow from query to answer

Important

All code examples and starter kits will be shared via the official repository after this workshop.

Technical Demo: Input Format

The official test set merges Type 1 and Type 2 queries into a unified stream. The system receives different fields depending on the query type.

Type 1 input — question + natural-language premises:

```
{
  "premises-NL": [
    "If a curriculum is well-structured ...",
    "..."
  ],
  "question": "Based on the premises, ..."
}
```

Type 2 input — question only:

```
{
  "question": "Calculate the energy stored in capacitor C ..."
}
```

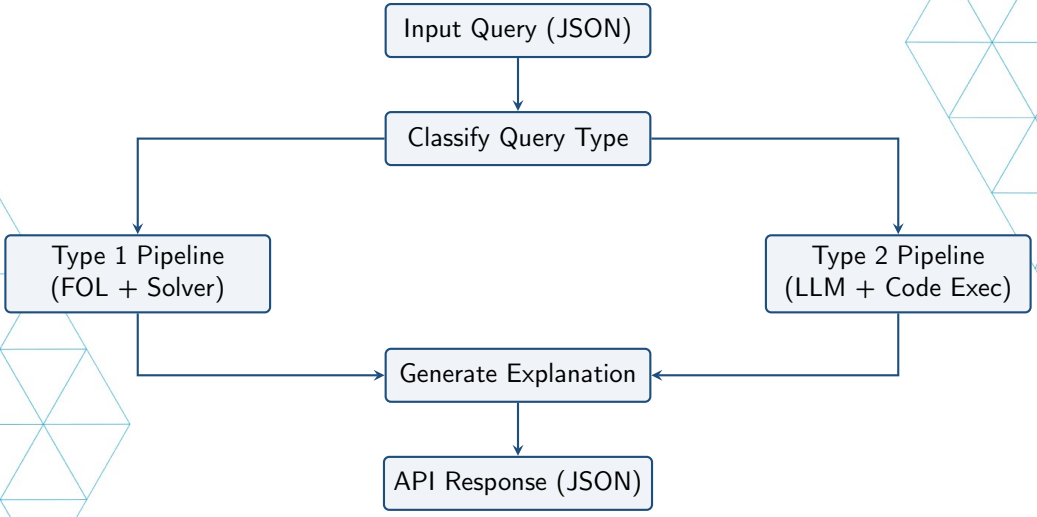
Technical Demo: Output Format

API Response (JSON) — only `answer` and `explanation` are mandatory; the optional fields contribute to the Reasoning Depth (P3) score.

```
{
  // Required
  "answer": "B",
  "explanation": "The voltage across R2 is calculated using ...",

  // Optional (encouraged, boosts P3)
  "fol": "ForAll(x, Resistor(x) -> HasVoltage(x, V))",
  "cot": [
    "Step 1: Identify the circuit topology ...",
    "Step 2: Apply Kirchhoff's voltage law ...",
    "Step 3: Solve for the unknown voltage ..."
  ],
  "premises": [
    "Ohm's law: V = IR",
    "KVL: sum of voltages in a loop = 0"
  ],
  "confidence": 0.92
}
```

Technical Demo: Baseline Pipeline



Thank You!

Good luck to all participants

https://ura.hcmut.edu.vn/exact

ura.hcmut@gmail.com

EXACT 2026 Organizing Committee